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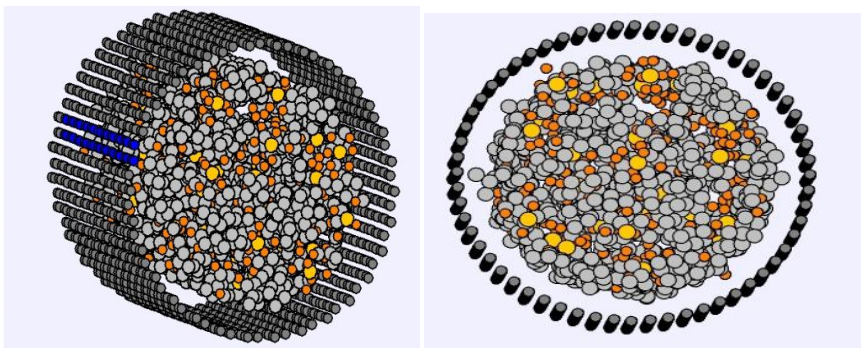
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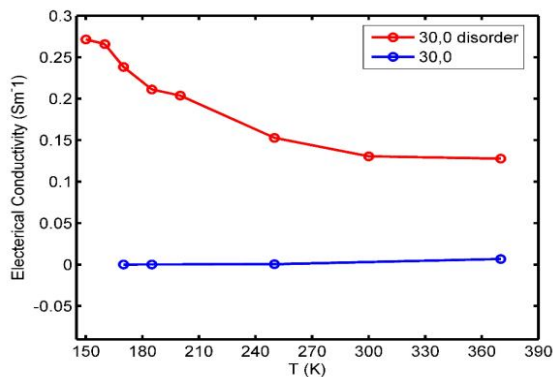
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Schematic presentation of [EMIM][PF6] encapsulated in zigzag type of carbon nanotube in presence nitrogen doping.



Electrical conductivity of [EMIM][PF6] increases versus the temperature when it encapsulated in CNT however electrical conductivity trend is reversed with temperature when nitrogen doped in zigzag CNT. In presence of impurity in CNT, ion transfer has been done via hydrogen bonding mechanism however in absence impurity; ion transfer has been done via diffusion mechanism in which cation has significant effect on ion transfer and electrical conductivity.

**The Effect Of Nitrogen Doping On Glass Transition and Electrical Conductivity Of
[EMIM][PF6] Ionic Liquid Encapsulated In Zigzag Carbon Nanotube**

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Abstract

Molecular level understanding of the properties of ionic liquids inside nanopores is needed in order to use ionic liquids for many applications such as electrolytes for energy storage in electric double-layer capacitors and dye-sensitized solar cells for conversion of solar energy. In this study classical molecular dynamics (MD) simulations have been performed to investigate the radial distribution, glass transition, ionic transfer number, and electrical conductivity of the ionic liquid 1-ethyl-3-methylimidazolium hexafluorophosphate [EMIM][PF₆] ionic liquid encapsulated in carbon nanotube (CNT). The effect of nitrogen as a doping element in CNT on these properties of [EMIM][PF₆] was also investigated by MD simulation, and the configurational entropy of [EMIM][PF₆] encapsulated in CNT was calculated in absence and presence of nitrogen as a doping element. The configurational entropy of [EMIM][PF₆] encapsulated in CNT is non-monotonic versus temperature in both the presence and absence of nitrogen doping. The glass transition of [EMIM][PF₆] encapsulated in CNT is shifted to high temperature with doped nitrogen. The Green-Kubo formalism was used to calculate the ionic transfer number of [EMIM][PF₆] encapsulated in CNT. Ionic transfer numbers show a maximum peak for cation transfer and a minimum peak for anion transfer with temperature. Electrical conductivity of [EMIM][PF₆] encapsulated in CNT decreases with increasing temperature in presence of doped nitrogen and increases in absence of nitrogen. The cationic conductivity also increases with temperature in the presence vs. absence of nitrogen doping. The MD findings for electrical conductivity and glass transition with temperature are in good agreement with available experimental data. The MD data shed new light on the effect of nitrogen doping on the mechanism of ion transfer. In presence of nitrogen ion transfer uses a hydrogen bonding mechanism, and in its absence ion transfer uses a diffusion mechanism in which the cation has a significant effect on ion transfer and electrical conductivity.

Keyword: Electrical Conductivity, Ionic Transference Number, Glass Transition, Configurational Entropy, Molecular Dynamics Simulation, Hydrogen Bonding

1. INTRODUCTION

Recently the interest in ionic liquids (ILs) has been grown rapidly due to their desirable properties and various applications in many fields.¹⁻³ ILs have been applied in many utilities such as green solvents for chemical synthesis, catalysis separations science as electrolytes for electrochemistry and photovoltaic applications.⁴⁻⁷ A fundamental understanding of the properties of ILs inside the nanopores is connected to their applications as electrolytes for energy storage in electric double-layer capacitors (EDLCs).⁸⁻¹⁴ dye-sensitized solar cells (DSSCs) for conversion of solar energy and separation of gases.¹⁵⁻¹⁷ Molecular dynamics (MD) simulations have been used for calculations of anomalous dependence of the surface tension with concentration for aqueous solutions 1-butyl-3-methylimidazolium tetrafluoroborate and 1-butyl-3-methylimidazolium hexafluorophosphate ionic liquids.¹⁸ Effect of water on surface structure of 1-butyl-1-methylpyrrolidinium trifluoromethyl sulfonylimide is considered via X-ray reflectometry. Presence of water in the mentioned ionic liquid has a non-monotonic effect on the overall thickness of the surface.¹⁹ Investigation of surface tension has been extended for aqueous solutions of magnesium acetate and magnesium nitrate at various concentrations.²⁰ X-ray photoelectron spectroscopy has been used for surface analysis of ionic liquids with and without lithium salt for application in lithium ion batteries.²¹

Novel amorphous ionic conductors with low glass transition temperature have been designed and synthesized for application in solid-state dye-sensitized solar cell (DSSCs or ssDSSCs).²² Highly inherent functionality^{4,5,7} and non-toxic is one of the most essential properties of synthesis ILs encapsulated in nanomaterials²³⁻²⁸ in comparison with other traditional organic compounds.²⁹⁻³⁴ Good mechanical and electronical properties can be achieved by mixing carbon nanotubes with imidazolium based ILs.³⁵⁻³⁶ Consequently, high electrical conductivity values for CNT can be expected for these materials in circuit performances.³⁷ Carbon nanotube based on IL has been applied as a new modified electrode to yield more sensitivity and stability. Melting point of CNTs increases with the increase of radius of CNTs and sensitivity of melting temperature to the radius is great in low radius.^{4,38-39} On the basis of previous study, ILs inside CNT has unusual thermal stability⁴⁰. Melting point and thermal decomposition temperature of several ILs increase by more than 200°C when they are confined inside multiwall carbon nanotubes (MWCNTs).^{35,41} In contrast, when confined inside silica nano-pores, melting points of IL decrease significantly.⁴²⁻⁴³ Previous survey shown that glass transition temperature (T_g) decrease for IL capsule within CNT

however the variation of melting temperature for IL is too small.⁴⁴ There is a jump for self-diffusion coefficient in the solid-liquid phase transition due to different atom mean free path.⁴⁵⁻⁴⁷ Diameter of CNT and pore loading of IL inside CNT has influence on structural and dynamical properties of ILs. Dynamics of bulk ionic liquids is faster than confined ionic liquid in CNT. Mean square displacement (MSD) of ion as a fluctuation around mean value of displacement for ion could be used for interpretation dynamical properties of ion such as diffusion coefficient in ionic liquids.

MSD for dynamics of the confined ions are heterogeneous and depend strongly on the distance of the ions from the pore walls.⁴¹⁻⁴² However, pore loading does not have any influence on diffusion coefficient and relaxation time of cations and anions within center of CNT in comparison with bulk IL.⁴⁸ Previous literature shows that the radial distribution function $g(r)$ of the ions in the center of the pore is similar to those of the bulk ILs but cations near to the pore walls arrange with their imidazolium ring parallel to the surface. Owing to fact available surface in CNT enhances with increasing CNT radius. High available surface area of CNT is the other factor which can influence the ordering of cations.⁴⁹ Result of MSD shows that self-diffusion coefficient of cations in IL is greater than these of anions.⁴⁹⁻⁵¹ IL confined in CNT shows faster dynamics and smaller relaxation time.⁴⁹ It was found that below the melting point, the ions within the nanopores were almost frozen around their positions. The average number of hydrogen bonds for each confined anion began to decline linearly by increasing the temperature above melting point. This decline led to a dramatic change in the packing arrangement of the confined ions and increases the ionic diffusivity of IL.⁵² Nanoparticle has significant effect in cation and anion arrangement and enhancement of heat transfer of liquid such as water and ionic liquid.⁵³⁻⁵⁵ On the basis of previous literature optical conductivity of zigzag CNT shows that electron-phonon interaction and temperature are essential factors on decreasing of optical conductivity.⁵⁶

Fundamental understanding of the structural and dynamical properties of IL inside nanopores is still lacking and such knowledge is crucial for the rational design of EDLCs, DSSCs and ion gels.

For instance, the application of IL-wrapped CNT gels for electrochemical biosensors, taking advantage of the synergetic catalysis effect between the high electrical conductivity of CNTs and the good electrical conductivity of ILs which display better performances than CNT-modified electrodes without ILs.⁵⁷ Moreover, radius of carbon nanotube has significant influence on structure properties of IL encapsulated CNT.⁵⁸

An accurate prediction of the electrical conductivity of ILs can be useful in many applications, such as batteries, fuel cells, double-layer capacitors and dye-sensitized solar cells⁵⁹⁻⁶⁰. Electrical conductivity of IL increases with increasing temperature.⁶¹⁻⁶² Similarly, by increasing the length of alkyl chain of the cation, increasing of the viscosity of the liquid is observed however, electrical conductivity, and glass transition temperature are decreased.⁶³

The electronic properties of the nanotube are determined by the tube chirality.⁶⁴ Similar to conventional semiconductor technology, the electronic properties of nanotubes can be tailored by introducing impurities. To this end, it has been suggested to dope carbon nanotubes with B and/or N atoms.⁶⁵⁻⁷³ Doping CNTs with nitrogen⁷⁴ is a promising way for tailoring their properties for applications in field emission⁷⁵, nano-electronics⁷⁶, gas sensing⁷⁷ or catalysis.⁷⁸

On the basis of previous literature low-energy ion implantation is a promising way to achieve a high fraction of substitutional B and N dopants in a carbon nanotube atomic network.⁷⁹

Zigzag types of carbon nanotube (30,0) as a semiconductor device is chosen because residual of 30 divide to 3 is zero and increase of diameter of zigzag from (35,0) and (40,0) leads to decreases the band gap. On the other expression semiconductors with a band gap that inversely depends on the nanotube diameter. As a result, there is semiconductor trend to the metallic trend for carbon nanotube as a nano-channel for investigation of thermodynamics and transport properties of [EMIM][PF₆] Ionic Liquid.⁸⁰

In this paper the effect of impurity and radius of CNT on glass transition and electrical conductivity of [EMIM][PF₆] encapsulated in CNT have been studied by using classical molecular dynamic simulations. We try to calculate hydrogen bonding namely the hydrogen bonds between the cations and anions, transference number of cation and anion to correlate these molecular level information to their contributions in electrical conductivity of [EMIM][PF₆] encapsulated in absence or presence nitrogen impurity in CNT. Calculation on configurational entropy is extended to get glass transition of [EMIM][PF₆] ionic liquid encapsulated in absence or presence nitrogen impurity in CNT. Fig.1.a. shows structure of 1-ethyl-3-methylimidazolium hexafluorophosphate ionic liquids.

2. COMPUTATIONAL DETAILS

Molecular dynamics simulations with canonical (NVT) ensemble have been performed for investigation of thermodynamic properties of IL.⁸¹ All atoms based optimized potentials for

liquid simulations (OPLS-AA) force field for ionic liquids which is developed by Lopes et al.⁸²⁻⁸³ with some modification⁸⁴ has been used for [EMIM][PF₆] ionic liquid. OPLS-AA force field results for nanostructures of confined ionic liquid have been reported in literature.^{48,85-89} In current study, static and dynamical properties of IL are obtained via force field within reference.⁹⁰ DL_POLY package version 4.03⁹¹ is applied for molecular dynamics simulations with NPT ensembles for room temperature of IL.

The time step of the simulations was 1 fs. MD simulation has been performed with 5×10^5 time steps with 400 000 equilibration steps. The simulations were performed with the Nosé–Hoover thermostat/barostat algorithm⁹¹⁻⁹² as implement in the DL_POLY program.⁹¹ The relaxation times which is used for the thermostat and barostat are 0.1 and 2.0 ps, respectively. The initial extensive MD simulations for pure ionic liquid were performed in the *NPT* ensemble for $p=1$ atm and $T=500$ K and annealing technique and cooling step with 50 K then using output structure for lower temperature until the total energy of the system converged. After that simulations were performed at the temperature range of 150K-370K. All interatomic interactions between the atoms in the simulation box have been calculated within the cutoff distance of $12 \text{ \AA}$. Molecular dynamics simulation has been done on 64 ion pairs of ionic liquid and temperature range 150K-370K. The Coulombic interactions were evaluated through the Ewald method using the convergence parameter and the largest reciprocal space vector that yielded a relative accuracy of 10^{-4} in the electrostatic energy. This part of the simulations has been used as optimization step of simulations of ionic liquid.

Time step of 1fs for molecular dynamics simulation for IL confined in CNT has been used by using both NPT and NVT ensembles. In the first step, ionic liquid is optimized then in the second step optimized ionic liquid is filled on carbon nanotube in random way. For the present simulations annealing technique has been applied with starting ionic liquid structure within CNT at upper temperature 500 K and doing simulation up to one nanosecond with cooling step 50 K and taking time 500 picoseconds (ps) with rate 0.1K/ps. Detail of simulation and optimization structure could be explained as following.

MD simulation with time step 1 fs has been performed by using optimized ionic liquid which is filled on carbon nanotube in random way with total time steps 5×10^5 and 400 000 equilibration steps by *NPT* ensemble. The end configuration of *NPT* ensemble was the beginning configuration of the next 5×10^5 step simulation. Afterwards, the MSDs and velocity autocorrelation function is calculated with *NVT* simulations each with a total 100 000 time step equilibration followed by 200 000 time steps for data collecting. The starting point of each *NVT* simulation and Evans thermostat⁹³ was the equilibrated by structure

configuration of the *NPT* simulation. By the end of the equilibration, the total energies, volumes and temperature are considered until the corresponding time series have been obtained stationary.

For non-linear molecule and in translational case, the simple leap frog algorithm could not be used directly because quaternion derivatives depend not only angular velocity but also on quaternion. However, a modified Leap frog algorithm may be used.⁹² Verlet Leap Frog method has been applied for solving equation of motion with including columbic long range interaction. The size of the rectangular unit cell was approximately $35 \times 37 \times 40 \text{ \AA}$, and the 3D periodic boundary conditions were applied for pure ionic liquid and presence simulation. Simulation with annealing technique has been extended from 500 K with cooling temperature 25K with rate 0.05K/ps and it is shown that physical result is same as with annealing technique with cooling temperature 50 K.

The total interaction potential between IL and the carbon atoms of CNT was calculated by the site-site interaction method with the following form:

$$U_{IC} = 4\varepsilon_{IC} \sum_{i=1}^{N_I} \sum_{j=1}^{N_C} \left[\left(\frac{\sigma_{IC}}{r_{ij}} \right)^{12} - \left(\frac{\sigma_{IC}}{r_{ij}} \right)^6 \right] \quad (1)$$

where the cross-correlation LJ parameters σ_{IC} and ε_{IC} were determined by the Lorentz-Berthelot rules.⁹⁴

The van der Waals force is explained via Lennard-Jones (LJ) 6-12 potential utilizing the geometric mean for ε and the arithmetic mean for σ .

Different diameter of CNT yields different density and other statics and transport properties. Annealing technique was used for equilibrating all systems and makes them ready for the production run which was 1ns.

In this study N-doping in zigzag CNT has been performed with different diameters, randomly and homogeneously⁹⁵ in 3atom percent.

Confined ILs inside a nanometer size pore shows low dynamics as it is reported in the work of Singh et al.⁴⁸ Theoretical result for glass transition of pure [EMIM][PF6] reported at 186.28K⁹⁶ while experimental glass transition measured is 173K.⁹⁷ Previous literature shows that ionic liquid has negligible vapor pressure.⁹⁸

The CNT systems chosen in the present study includes zigzag CNTs with three different radii and two nitrogen doped zigzag CNTs with diameters ranging from 3 to 4 nm. Table 1 gives the number of carbon atoms, diameters (D), and lengths (L) for the all CNTs which is used in this work. The slit pore was filled with 64 IL ion pairs and MD simulations were performed at 150K-370K.

The average number of the hydrogen bonds between the confined IL molecules inside the CNTs has been calculated on the basis of hydrogen bond length of 2.58Å° between the H atoms from the cations and the F atoms from the anions and angle of CR(carbon ring)-H with F atom and angle for C2 (alkyl carbon)-H with F atom are around 140 ° and 70 ° respectively.

3. RESULTS AND DISCUSSION

Glass phase was formed by quickly cooling the viscous liquid below its melting point, avoiding crystallization and continuing to cool to the glass transition temperature (*T_g*).⁹⁹The glass transition temperature can be defined in several ways. The first definition of *T_g* is based on heat capacity which dramatically decreases upon glass formation. Configurational entropy, radial distribution function, self-diffusion coefficient, number of average hydrogen bonds and transference number are the other methods for recognition of glass transition *T_g*.

3.1. Glass temperature based on heat capacity. Heat capacity in constant volume (*C_v*) and constant pressure (*C_p*), are appropriated to study the glass transition in ionic liquids. Near the transition, heat capacity undergoes a first order phase transition.

In canonical ensemble heat capacity is related to the fluctuations in the potential energy

$$C_v = \frac{\langle E^2 \rangle - \langle E \rangle^2}{k_B T^2}$$
 (2)

where *E* and *k_B*, *T* are the potential energy, Boltzmann constant and temperature respectively. On the basis of Eq.2 the heat capacity could be calculated for the glass transition. The heat capacity determined as a function of temperature usually gives a single peak which attributes to glass transition temperature.

To identify the glass transition temperature, the specific heat capacity at constant volume must be calculated by the differential of internal energy with respect to temperature which is shown in Eq.(3)

$$C_v = \frac{dE(T)}{dT}$$
 (3)

Heat capacity at constant pressure can be calculated by the following equation.

$$C_p(T) = \left(\frac{dU_P}{dT}\right)_P + P \left(\frac{dV}{dT}\right)_P$$
 (4)

Heat capacity result for [EMIM][PF6] encapsulated in CNT shows that in glass transition temperature the heat capacity decreases suddenly. Our MD simulation estimates glass transition for pure [EMIM][PF6] ionic liquid is 167.5 K which shows 3% percent error compare to available experimental data.

Our computational results regarding [EMIM][PF6] ionic liquid encapsulated in CNT(30,0) show that glass transition occurs at the temperature of 187.5 K while experimental glass transition temperature for pure bulk [EMIM][PF6] is reported to be 173K.⁹⁷ On the basis of MD result glass transition temperature for confined IL in CNT increases because more thermal stability could be yielded with strength interaction between the IL and CNT wall.

Result of glass transition of pure bulk [EMIM][PF6] has been shown at Figure.1.b.

The effect of carbon nanotubes radius on the heat capacity for the glass transition temperature has been shown in Figure 1.c. On the basis of Fig.1.c. an increase of the radius of CNT lead to a decrease of heat capacity significantly because the space for rearrangement of structures increases and irregularity for structures intensified.

It could be imagined when [EMIM][PF6] encapsulated in CNT(30,0) immersed in [EMIM][PF6] ionic liquid, then it is a solvated [EMIM][PF6] encapsulated in CNT(30,0).

Calculation of glass transition has been done for solvated [EMIM][PF6] encapsulated in CNT(30,0) and it occurs at 172.5K which is lower than glass transition 187.5 K of [EMIM][PF6] ionic liquid encapsulated in CNT(30,0). Result of glass transition of solvated [EMIM][PF6] encapsulated in CNT(30,0) in [EMIM][PF6] is presented at Fig.1.d.

Snapshots from MD simulations for ionic liquid encapsulated in CNT for (40,0) and (30,0) before and after glass transition have been shown in Fig.2a,2b and Fig.3a,3b respectively. Structure of ionic liquid encapsulated in (40,0) before glass temperature is more different than after glass temperature but for (30,0) structure of ionic liquid confined in CNT before and after glass temperature are similar. Radial distribution function of the carbon of nanotube and carbon of the cation ring (CR-Cr), Fig.4., for ionic liquid encapsulated in (30,0) show that the most of main peak before and after glass transition will be unchanged however for ionic liquid encapsulated in (40,0) CNT, radial distribution peak shape changes completely. As a result, structure of ionic liquid encapsulated in CNT (40,0) becomes more disturbed than the structure of ionic liquid encapsulated in CNT(30,0).

Result of radial distribution function $g(r)$ of cation around carbon wall for [EMIM][PF6] encapsulated in (30,0) and (40,0) has been presented at Figure.4. According to the $g(r)$ at the glass temperature, number of cation near the CNT wall in (30,0) is more than those for (40,0) CNT.

Less population of cation near the CNT wall for stability of structure will be resulted for small heat capacity of CNT (40,0) in comparison with CNT (30,0) in glass transition.

Heat capacity at constant volume C_v and pressure C_p for carbon nanotube (30,0) and (35,0) in presence and absence of nitrogen impurities has been shown in Fig.5 and Fig.6.a respectively.

In (35,0) CNT, at glass transition temperature decrease of heat capacity in presence of impurities is more than those for the lack of impurities. In (30,0) CNT, at glass transition point, a jump in heat capacity in presence of impurities is more significant than those in the lack of impurities as well. When ionic liquid encapsulated in CNT in presence of impurities, for (30,0) CNT structural rearrangement of ionic liquid within CNT is harder in comparison with (35,0) CNT due to small radius. As a result, changing configurational energy near the glass transition temperature for (30,0) CNT is more than (35,0) CNT.

Configuration energy as an interaction energy due to stabilization energy between two[EMIM][PF6] encapsulated in CNT(30,0) (E2) in comparison with two single [EMIM][PF6] encapsulated in CNT(30,0) (E1) has been calculated versus temperatures and the results have been shown in Figure.6.b. On the basis of Fig.6.b stabilization energy of two ionic liquid encapsulated in CNT in comparison with two single ionic liquid encapsulated in CNT has complicated trend as a function of temperature. Generally, before glass transition stabilization energy is more than after glass transition and in glass transition there is one maximum for instability for interaction of two ionic liquid encapsulated in CNT. Heat capacity difference between CNT(35,0) and (30,0) is related to the entropy and energy effects. Structure of confined IL in CNT for (35,0) case gets more entropy and more amorphous structure in comparison with (30,0).

Structure of ionic liquid inside of CNT in presence of impurities before and after glass transition for CNT (30,0) and CNT(35,0) is presented in Fig.7 and Fig.8 respectively. Result of radial distribution function between (Cr-Cr) in absence and presence of impurity for CNT (30,0), (35,0) is presented at Figure.9. On the basis of Fig.9 impurities have a major effect on the structure of ionic liquid inside big radius of CNT in the glass transition point however impurity has negligible effect on the structure of ionic liquid encapsulated with small CNT radius.

Another reason for sharp increase of heat capacity at glass transition temperature for IL encapsulated in (30,0) CNT in presence of impurity is the average number of hydrogen bonds, see Fig 10.

The average number of hydrogen bonding near the glass transition in presence of doping decreases and less stability between F and N atom with hydrogen atom yields. Increase the temperature can disturb the structure of ionic liquid with less configuration energy and hydrogen bonding gets more jumps in heat capacity near the glass transition in comparison with absence of impurity.

3.1. Glass transition temperature determination from Configurational Entropy.

The specific entropy, S , of an ionic liquid, even though it is not the state of low free energy, can be calculated in the usual way by integrating over the measured specific heat. Configurational entropy can be calculated as a following equation¹⁰⁰

$$\Delta S(T) = \int_{T_1}^{T_2} \frac{C_P(T)}{T} dT \quad (5)$$

The crystal entropy at the glass transition temperature T_g can be calculated if the specific heat of the crystal is measured from T_0 to T_g .

MD result shows that entropy increases with increasing temperature, but this increase is much more in the glass transition temperature. Diagram for configurational entropy versus temperature for two different radiuses of CNTs have been shown in Figure 11. When radius of CNT increases, the amount of configurational entropy increases as well.

Diagram for configurational entropy versus temperature for two types of CNTs in present of nitrogen impurity shows in Figure 12.

According to Fig. 12, with nitrogen doping in CNT, the amount of entropy for ILs encapsulated in CNTs increases.

Figure 9 shows the effect of nitrogen impurity in (30,0) and (35,0) CNT on the $g(r)$ values near the glass transition temperature. In (35,0) with presence of impurities, the peak of radial distribution function decreases due to decreases of the number of interactions between the carbon atoms of CNT with carbon atoms of imidazolium ring. But in the case of (30,0) the presence of impurities, the first peak of the radial distribution function increases in short distance and in long distance similar behavior can be found with lack of impurities. In (30,0) CNT, the structure of ILs completely packed inside the CNT so that the number of interaction between imidazolium ring and carbon atoms of CNT wall decreases and the number of cation with orientation near to CNT wall decreases thus entropy increases.

3.3. Discussion on the radial distribution function (RDF), $g(r)$. The long-range correlations in the solid phase is obvious. The $g(r)$, for the various atom pairs i and j is given by

$$g(r) = \frac{1}{\rho N} \langle \sum_i \sum_j \delta[r - r_{ij}] \rangle \quad (6)$$

where N is the total number of atoms, $\rho = N/V$ is the number density, r_{ij} is the distance between atoms i and j , and the brackets indicate the ensemble average. The RDF alone is not very sensitive measure to reproduce accurately all the physical properties of materials. However, calculation of other thermodynamic properties such as number of hydrogen bonds can give useful information about the structural properties of the phases.

Figure 13 shows the $g(r)$ for the C_r carbon of cation and carbon of CNT(C-R) for (30,0),(40,0) in glass temperatures. On the basis of Fig.13 big radius for zigzag CNT makes more intensity peak for radial distribution function between C_r and C-R.

On the basis of Fig.4, peak intensity of radial distribution function decreases with increasing the temperature. For (30,0), the intensity of radial distribution function peak between C_r and C-R at 370 K is greater than 300 K, but in 185 K near the glass transition temperature the peak intensity decreases significantly in comparison with other temperatures due to irregularity in the crystal lattice in the glass transition temperature.

3.4. Investigation of the average number of hydrogen bonds in IL encapsulated in CNT.

The average number of the hydrogen bonds between the confined IL molecules inside the CNTs have been obtained by using the calculation of the hydrogen bonds length ($2.58\text{\AA}$) between the H atoms of the cations and the F atoms of the anions and the angle between CR (carbon of ring)-H with F atom and angle of C2 (carbon of alkyl group)-H with F atom to be 140° and 70° respectively.

Radius effect has been considered on the number of hydrogen bonding versus temperature and the result is presented at Fig.14. According to the Fig.14 number of hydrogen bonds decreases with the increase of the radius of CNT.

By increasing the radius of CNT, ionic liquid molecules have more space to move and then they can increase their entropy instead of configurational energy. As a result, ionic liquid encapsulated in CNT has small chance for formation of hydrogen bonds in big radius of CNT due to increasing of the entropy.

This can be due to the fact that as the amplitude of the ionic vibration increases with temperature⁵⁴⁻⁵⁵, the average number of hydrogen bonds declines linearly. This decline might be directly responsible for the dramatic change in packing arrangement of ions inside the CNT. These changes in the molecular packing also affect the ionic diffusion behavior.¹⁰¹

At glass temperature in zigzagCNT (30,0) and (40,0), average number of hydrogen bonds decreases. This can be due to the fact that as the entropy of the ionic liquid increases at glass transition temperature due to decreases of the strength of the interaction between the ions.

In Figure 10 result of the average number of hydrogen bonds for (30,0) and (35,0) CNT versus temperature has been shown in present and lack of nitrogen impurities.

By comparing the two graphs, we find that impurities have a different effect on the two radius of CNTs. In (35,0) CNT, average number of hydrogen bonds between cations and anions of [EMIM][PF₆] in present of impurities in most of temperatures increases, however in glass transition temperature the average number of hydrogen bonding decreases in comparison with absence of nitrogen impurity.

But the behavior of ILs inside of (30,0) CNT is completely different. Average number of hydrogen bonds for (30,0) at glass transition temperature increases suddenly. This behavior is appropriated with stability of the structure of the ions of ionic liquid encapsulated in (30,0)CNT and consistent with increases of heat capacity of ionic liquid encapsulated inside of (30,0) CNT in present of impurities. Snapshots from MD simulations of ILs in (30,0)CNT at 160K, 185K, 370K have been presented in Fig. 15.

3.5. Calculation of transference number. Thermal energy makes a translational motion for pure liquid in thermodynamic equilibrium and the translational motion is called self-diffusion.¹⁰² It is demonstrated that the diffusion coefficients calculated from the Green-Kubo relation are very sensitive to the selected upper time limits of velocity autocorrelation function (VACF) integrals. It seems that the diffusion results from the Green-Kubo relation are closer to the experimental values than the results from the Einstein relation¹⁰³.

For this reason, we determined self-diffusion coefficient by Green-Kubo relation¹⁰⁴

$$D = \frac{1}{3N_m} \int_0^\infty \left\langle \sum_{j=1}^{N_m} v_j(t) \cdot v_j(0) \right\rangle dt \quad (7)$$

where the $\left\langle \sum_{j=1}^{N_m} v_j(t) \cdot v_j(0) \right\rangle$ in the Green-Kubo relation is usually called velocity autocorrelation function.

Ionic liquids can be used as electrolytes in electrochemical processes and devices.

In many electrolytes total conductivity as well as charged distribution of species are essential parameters for charge transport.¹⁰⁵

For simple 1:1 electrolytes, MX consisting of monovalent ions M^+ and X^- , such as the ionic liquids the transference numbers may be estimated from the diffusion coefficients of the cation and anion¹⁰³,

$$t_+ = \frac{D_+}{D_+ + D_-} \quad (8)$$

$$t_- = \frac{D_-}{D_+ + D_-} \quad (9)$$

By application Green-Kubo formalism and using Eq.7 self-diffusion coefficients of anions D_- and cations D_+ are calculated. In Eq.8 and Eq.9, t_+ and t_- are cation and anion transference number respectively.

Between cationic and anionic transference numbers these equations $t_+ + t_- = 1$ and $0 \leq t_i \leq 1$ are valid.

Figure 16 shows variation of transference number with temperature for three radii of zigzag CNTs. With increasing the temperature, t_+ increase and at the glass transition temperature t_+ is maximum, but with increasing the temperature, t_- decrease and at the glass transition temperature t_- is minimum. Trend of t_+ as a cation and t_- as an anion transference numbers versus temperature are opposite to each other.

3.6. Effect of Impurity on the Transference Number

Cationic transference number t_+ for ILs encapsulated in CNT in present of impurity has different behavior in different radius of CNT. Figure 17 shows the variation of transference number (t_+) with temperature for IL encapsulated in CNT (30, 0) and (35, 0) in present of nitrogen impurity in CNT and pure CNT. Except around glass transition temperature, (t_+) for ILs confined in (35,0) CNT in present of impurity is more than (t_+) for ILs confined in (35,0) CNT in pure CNT. However (t_+) for ILs confined in (30,0) CNT in present of impurity is less than (t_+) for ILs confined in (30,0) CNT in pure CNT. So that, with increases of the radius of CNT in present of impurity, the value of cationic transference number (t_+) increases. On the basis of Fig.17 by increasing the radius of CNT the number of hydrogen bonds between the cations and doped nitrogen atoms inside the CNT decreases therefore (t_+) increases.

Figure 18 shows variation of transference number (t_-) with temperature for IL encapsulated in CNT (30, 0) and (35, 0) in present of impurity and pure CNT.

(t_-) for ILs encapsulated in CNT in present of impurity have different behavior in different radii of CNT as well. Except around glass transition temperature, (t_-) for ILs confined in (35,0) CNT in present of impurity is less than (t_-) for ILs confined in (35,0)

CNT in pure CNT. For the most of temperatures except temperature around glass transition, (t_g) for ILs confined in (30,0) CNT in present of impurity is more than (t_g) for ILs confined in (30,0) CNT in pure CNT. Thus with increases of the radius of CNT in present of impurity (t_g) decreases. With increases of the radius of CNT, interaction between the F atoms of anions and doped nitrogen atoms of CNT decreases and repulsion between anions and nitrogen atoms of doped CNT decreases. Consequently, increasing of the radius of doped nitrogen atoms of CNT decreases the anion transference number.

3.7. The Evaluation of Electrical Conductivity

The electrical conductivity was calculated from both the Nernst–Einstein equation and from the slope of the collective mean square displacement (MSD) using Einstein relation:¹⁰⁶

$$D_s = \frac{1}{6} \lim_{t \rightarrow \infty} \frac{d}{dt} \left\langle \sum_{i=1}^N \left| r_i^{(c)}(t) - r_i^{(c)}(0) \right|^2 \right\rangle \quad (10)$$

The term $\sum_{i=1}^N \left| r_i^{(c)}(t) - r_i^{(c)}(0) \right|^2$ in the Einstein relation is regarded as mean square displacement.

The Nernst–Einstein equation is given by Equation¹⁰⁷⁻¹⁰⁸

$$Ec_{NE} = \frac{Ne^2}{VTk_B} (Z_-^2 D_- + Z_+^2 D_+) \quad (11)$$

where N , e , V , k_B , T , Z_- (Z_+) denote the number of ion pairs, elementary electronic charge, volume of the system, Boltzmann constant, temperature in Kelvin, formal charge on the anion (cation) respectively.

To calculate the true electrical conductivity of the system, one should consider the collective counterpart of the MSD and the corresponding Einstein equation [Eq. (12)]:^{107,109,110}

$$Ec = \frac{\lim_{t \rightarrow \infty} e^2}{6tVTk_B} \left\langle \sum_{i=1}^N \sum_{j=1}^N Z_i Z_j \left(\left[r_i^{(c)}(t) - r_i^{(c)}(0) \right] \cdot \left[r_j^{(c)}(t) - r_j^{(c)}(0) \right] \right) \right\rangle \quad (12)$$

Diagram for the electrical conductivity for bulk ILs as a function of temperature shows that the electrical conductivity increases with increases of the temperature (see Fig.19). At melting point temperature (333K) the electrical conductivity increases suddenly. After melting point of ILs the motion of ions is faster so that the electrical conductivity value in upper melting temperatures bigger than the temperature before the melting point.

In Figure 20, electrical conductivity (Ec) results from MD simulations are compared with experimental results at different temperatures. For [EMIM][PF6], the simulations reproduce experimental data rather well over the entire temperature range. The calculated results for [EMIM][PF6] are in good agreement with experimental results. Previous article shows that the

simulation results for electrical conductivity of different ILs is less than experimental results¹¹¹ and it is related to the nature of used force field which under estimate the conductivity. As a result of using force field our simulation results around 55% percent error smaller than the experimental results.

3.8. The Electrical Conductivity of ILs Confined in Pure CNT

To calculate the electrical conductivity for [EMIM][PF6] confined in CNT at the first step, self- diffusion coefficient has been calculated from Einstein relation from Eq. (10). Initially, the diagram MSD for cations and anions of IL as a function of time is plotted in Figure 21.a, and b respectively. According to MSD diagram we calculated electrical conductivity from Nernst–Einstein formula, Eq. (11).

Figure 22 shows the electrical conductivity as a function of temperature for [EMIM][PF6] inside of different radius of CNTs. According to Fig.22, electrical conductivity of ILs inside of CNT increases with increase of the radius of CNT. Result of $g(r)$ shows that the number of ions near the CNT wall decreased with increasing radius of CNTs which means [EMIM][PF6] inside of (40,0) CNT has more electrical conductivity than [EMIM][PF6] inside of (30,0) CNT.

In Figure 23, the electrical conductivity of [EMIM][PF6] confined in (40,0) CNT and pure [EMIM][PF6] is compared as a function of temperature.

Figure 23 shows that the electrical conductivity of [EMIM][PF6] confined in (40,0) CNT is less than pure [EMIM][PF6]. Cations and anions of ILs have interaction with CNT wall, so that, the ions move more slowly and electrical conductivity of [EMIM][PF6] decreases when it confined in CNT electrical conductivity of IL confined in (30,0)CNT in presence and absence of impurity has been presented at Fig.24. On the basis of Fig.24 electrical conductivity of IL confined in (30,0)CNT increases in presence of impurity. Figure 24 shows that electrical conductivity of IL confined in (30,0)CNT increases with temperature, however in presence of impurity the electrical conductivity value decreases as a function of temperature. It seems that in presence of impurity electrical conductivity of ionic liquid encapsulated in CNT occurs via hydrogen bonding mechanism. By forming of hydrogen bonding ion can transport with CNT and it has significant contribution for electrical conductivity. On the basis of Fig.6.a hydrogen bonding decreases with the temperature. As a result, electrical conductivity decreases versus temperature for ionic liquid encapsulated in CNT in presence of impurity. For ionic liquid encapsulated in CNT ion transport has been done via diffusion mechanism and cation has significant contribution in electrical

conductivity. On the basis of Figure 17 transfer number for cation increases as a function of temperature which confirms diffusion mechanism for ion transport.

4. Conclusion

Classical molecular dynamics simulations have been performed to investigate hydrogen bonding, electrical conductivity, ionic transference number and configurational entropy of [EMIM][PF₆] encapsulated in zigzag type of CNT. Molecular dynamics simulations revealed that radius and doping does not have significant influences on glass transition temperature of [EMIM][PF₆] encapsulated in zigzag type of CNT. Radius of zigzag carbon nanotube does not change the trend of configuration entropy of [EMIM][PF₆] encapsulated in zigzag type of CNT as a function of temperature however the value of configuration entropy increases versus temperature with the increase of the radius of zigzag CNT except around glass transition. There is a minimum value as a peak in absence of impurity for configurational entropy of [EMIM][PF₆] encapsulated in zigzag type of CNT and this minimum observed at glass transition temperature. Configuration entropy behavior in presence of nitrogen impurity in CNT near glass transition temperature is depends on radius of zigzag carbon nanotube. The peak of average number of hydrogen bonds around the glass transition temperature for [EMIM][PF₆] encapsulated in zigzag type of CNT with nitrogen doping in CNT become sharper in comparison to those in absence of nitrogen impurity. The trend of average number of hydrogen bonds between the cations and anions confined inside the smallest and largest radius of CNTs at different temperatures are approximately the same and the number of average hydrogen bonds of [EMIM][PF₆] encapsulated in zigzag type of CNT decrease with the increase zigzag CNT radius and temperature. Green Kubo formalism regarding ionic transfer shows that the trend of cation transference number is opposite to anionic transference number for [EMIM][PF₆] encapsulated in zigzag type of CNT versus temperatures. Electrical conductivity of [EMIM][PF₆] encapsulated in zigzag type of CNT increases with the radius of CNT. The value of electrical conductivity of [EMIM][PF₆] decreases when it is encapsulated in zigzag CNT. The trend of electrical conductivity of pure [EMIM][PF₆] and [EMIM][PF₆] encapsulated in zigzag type of CNT with temperature are the same and they increases with the temperature.

When nitrogen doped in zigzag CNT, electrical conductivity of [EMIM][PF₆] encapsulated in zigzag type of CNT increase in comparison with those in absence of nitrogen impurity and in presence of nitrogen doping in CNT, electrical conductivity decreases with the temperature. Radial distribution function shows that the number of carbon in cation ring with carbon wall of CNT in zigzag type decreases with increase of nanotube radius. On the basis of radial distribution function, in glass transition number of carbon in cation ring around carbon of nanotube wall decreases. At the glass transition nitrogen doping in CNT leads to diminish number of carbon in cation ring around carbon wall in CNT in comparison to those in absence of impurity. Hydrogen bonding has significant contribution in ion transfer of ionic liquid in CNT in presence of impurity but diffusion of cation has significant role for ion transfer of ionic liquid encapsulated CNT without impurity.

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Table.1

carbon nanotube (m, n)	(30,0)	(35,0)	(40,0)	(30,0) 3N	(35,0) 3N
carbon atoms	1440	1680	1920	1397	1630
D/Å	30	35	40	30	35
L/Å	52.03	49.07	50.1	51.64	48.62

Table Caption

Table 1: Simulation Cases for Carbon Nanotubes, where (m, n) are the Carbon Nanotube's Indices, D Is the Carbon Nanotube's Diameter, L Is the Carbon Nanotube's Length and N is the disorder nitrogen.

Figure caption

Fig.1.a. Structure representation 1-Ethyl-3-methylimidazolium Hexafluorophosphate Ionic Liquid, symbol: blue: Nitrogen, gray: carbon, white: Hydrogen, Orange: Phosphorus, Pale blue: Fluorine, b). Specific heat capacity vs temperature for pure [EMIM][PF6] ionic liquid due to determination of glass transition. c). Specific heat capacity vs temperature for [EMIM][PF6] encapsulated in CNT with different radiuses at constant volume and pressure. d). Specific heat capacity vs temperature for solvated [EMIM][PF6] encapsulated in CNT (30,0) for finding glass transition.

Fig .2. Two snapshots of [EMIM][PF6] encapsulated in CNT in zigzag (40,0) at 160K(left) (a) and 185K(right) (b) temperature

Fig .3. Two snapshots of [EMIM][PF6] encapsulated in CNT in zigzag (30,0) at a) 160K(left) b) 185K(right) temperature

Fig .4. Radial distribution function $g(r)$ carbon cation (Cr) and carbon CNT(C-R) for (30,0),(40,0) in different temperature.

Fig.5.Heat Capacity (C_p and C_v) graphs for carbon nanotube (35,0) in the presence of impurities and the lack of impurities.

Fig.6.a).Shown heat Capacity (C_p and C_v) graphs for carbon nanotube (30,0) in the presence of impurities and the lack of impurities. b) .Energy difference between interaction of two [EMIM][PF6] encapsulated in CNT from twice energy of single [EMIM][PF6] encapsulated in CNT versus temperature

Fig 7. Two snapshots of [EMIM][PF6] encapsulated in CNT in zigzag (35,0) in presence of impurities at 160K(left) and 185K(right) temperature

Fig.8. Two snapshots of [EMIM][PF6] encapsulated in CNT in zigzag (30,0) in presence of impurities at 160K(left) and 185K(right) temperature

Fig.9 .Radial distribution function $g(r)$ carbon cation (Cr) and carbon CNT(C-R) for (30,0),(35,0) at glass temperature in present of impurity and pure.

Fig 10. The average number of hydrogen bonds between the cations and anions confined into the (30,0) and (35,0) CNTs at different temperatures in present of impurity and pure.

Fig 11. Compare configurational entropy for IL encapsulated in different radius CNT.

Fig 12. Compare configurational entropy for IL encapsulated in different radius CNT in present of impurity and pure.

Fig 13.Radial distribution function $g(r)$ carbon cation (Cr) and carbon CNT(C-R) for (30,0) up diagram,(40,0) down diagram, at glass temperature.

Fig14.The average number of hydrogen bonds between the cations and anions confined into the smallest and largest CNTs radius versus temperatures.

Fig. 15.Three snapshots of [EMIM][PF6] encapsulated in zigzag(30,0) CNT in three temperature 160 K, 185K and 370 K.

Fig. 16.Variation of transference numbers with temperature for three radiuses of zigzag CNTs.

Fig. 17. Variation of transference number (t^+)with temperature for IL encapsulated in a) CNT (30, 0) b) (35, 0) in present impurity and pure CNT.

Fig. 18.Variation of transference number (t^-) with temperature for IL encapsulated in CNT (30, 0) and (35, 0) in present impurity and pure CNT.

Fig.19. The electrical conductivity for bulk ILs as a function of temperature

Fig.20.Plot the experiment and simulation of $\ln(Ec)$ versus $100 * \frac{1}{T-T_g}$ for bulk ILs

Fig. 21. Diagram MSD versus time(s) and fitting data for [EMIM][PF6] encapsulated in (40,0) CNT at 200K.

Fig.22. The electrical conductivity due to ion contribution as a function of temperature For [EMIM][PF6] inside of different radius CNT.

Fig.23. Compared the electrical conductivity due to ion contribution as a function of temperature For [EMIM][PF6]confined in (40,0) CNT and pure [EMIM][PF6].

Fig.24. Electrical conductivity due to ion contribution of IL confined in (30,0) CNT in presence and absence of impurity

Fig.1.a.

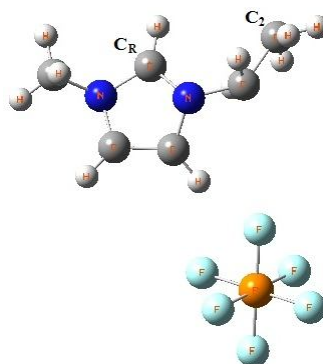


Fig.1.b.

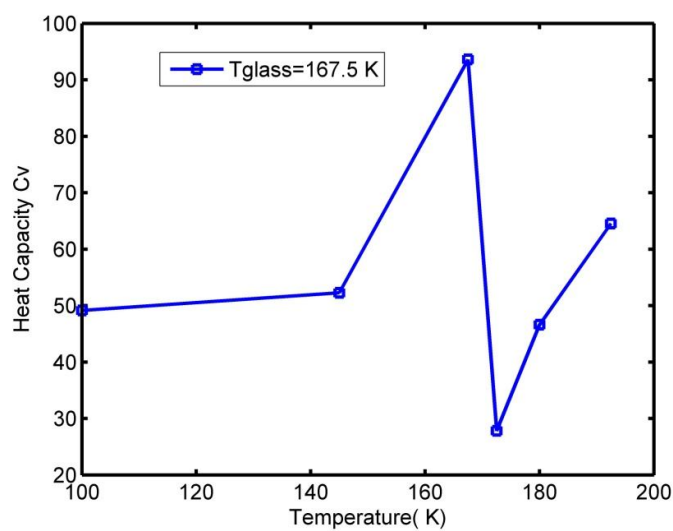


Fig.1.c.

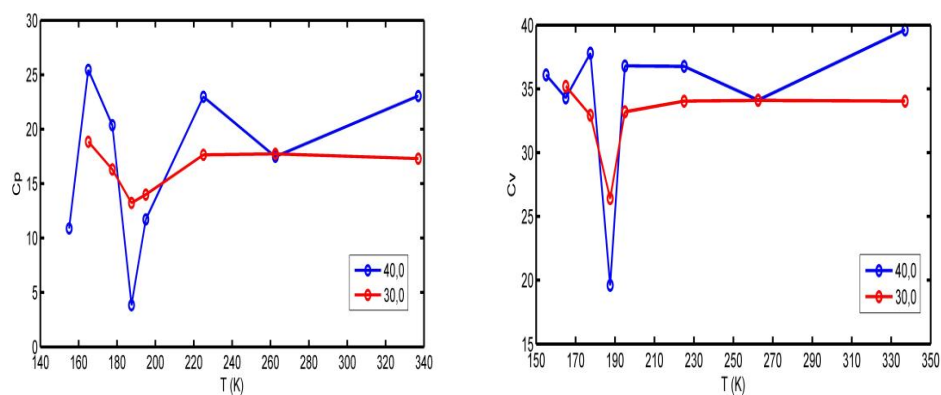


Fig.1.d.

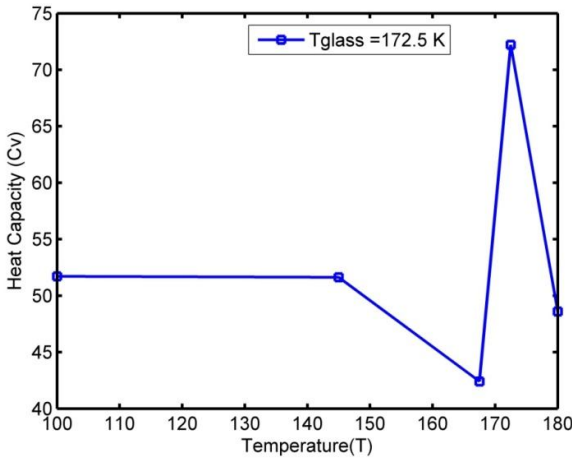


Fig.2.

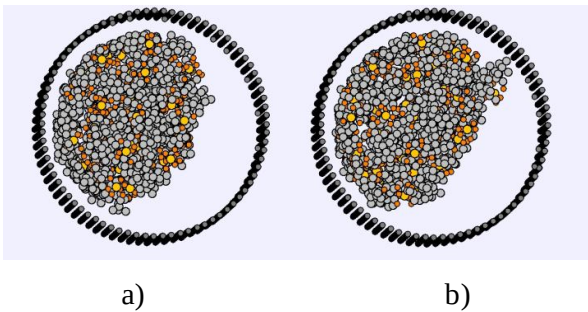


Fig.3

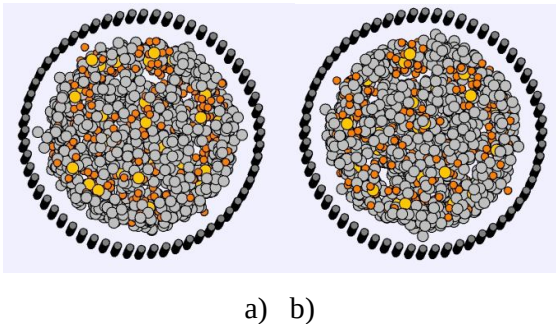


Fig.4

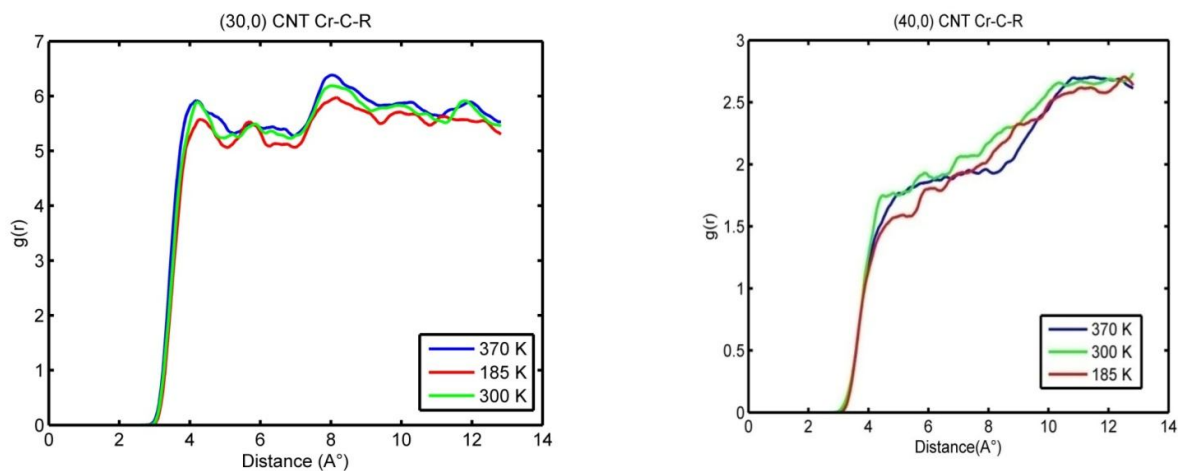


Fig.5.

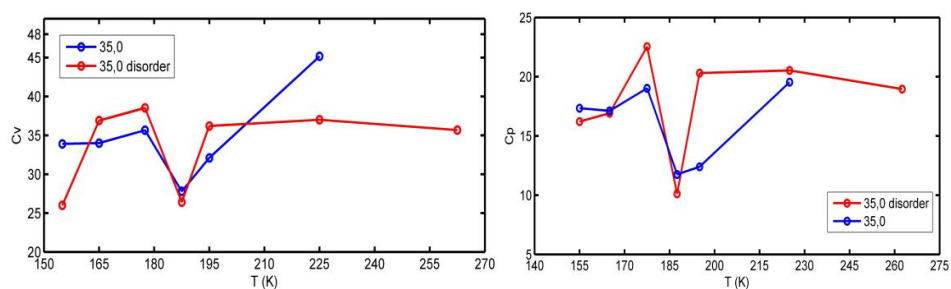


Fig.6.a.

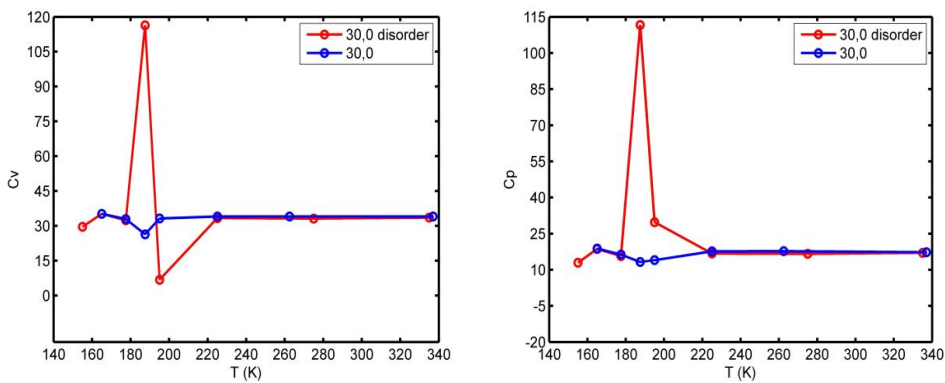


Fig.6.b.

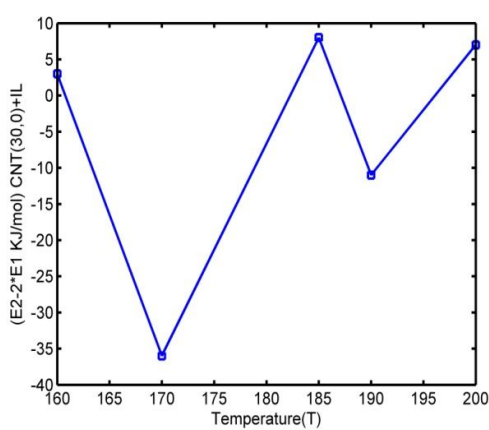


Fig.7.

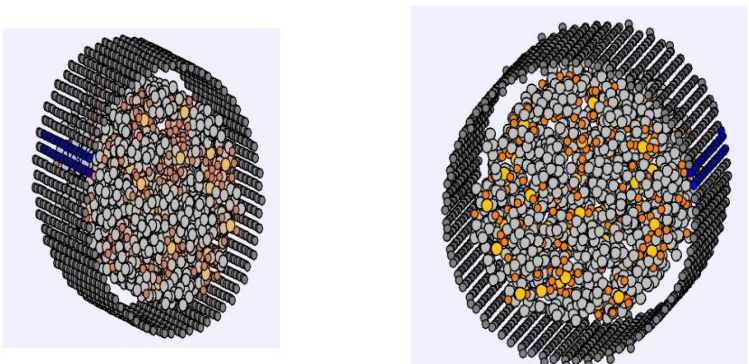


Fig.8

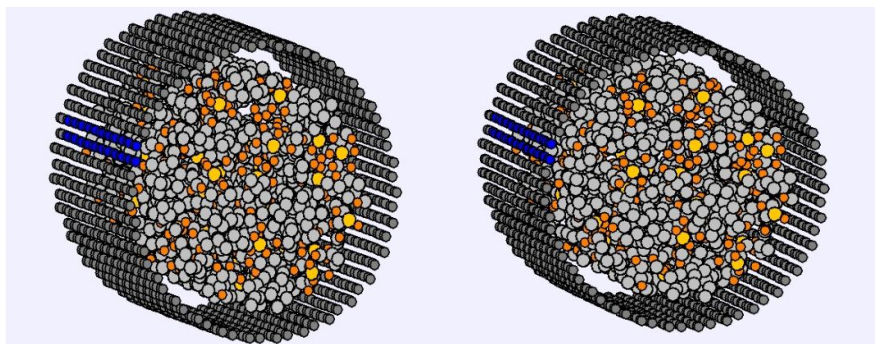


Fig.9

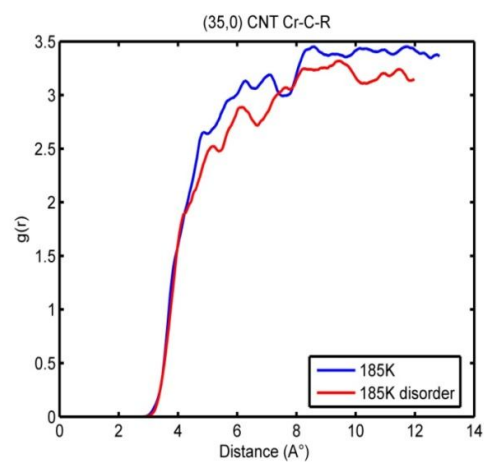
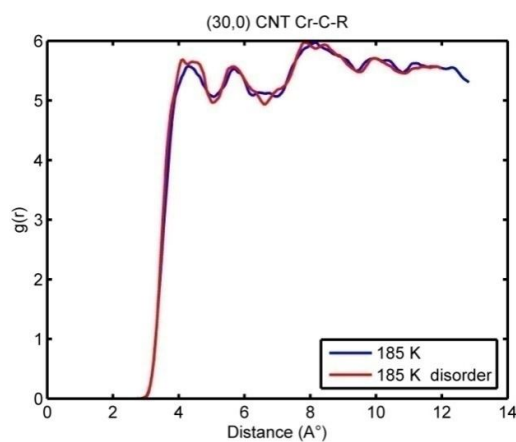


Fig.10

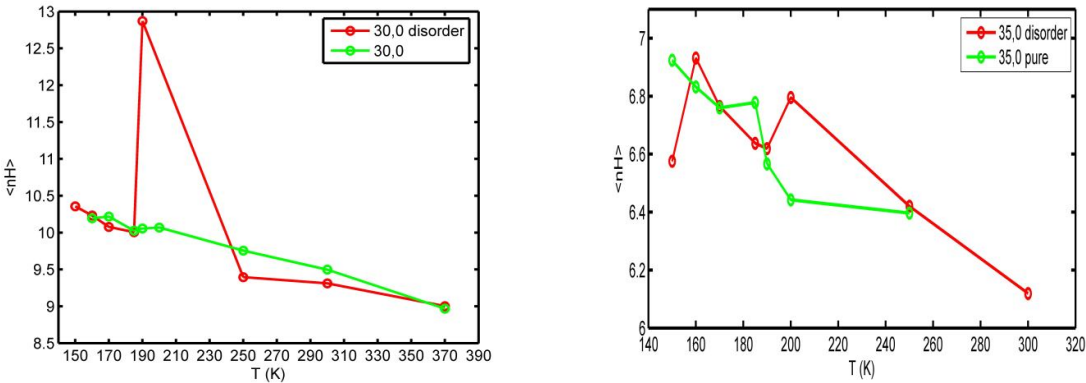


Fig.11

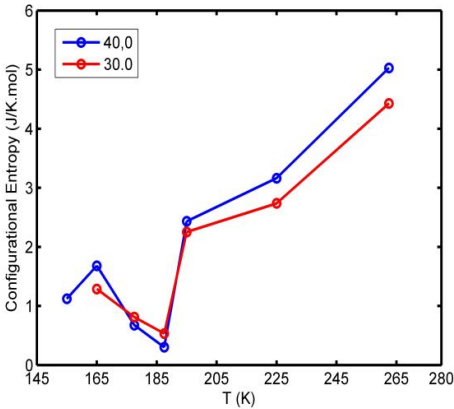


Fig.12

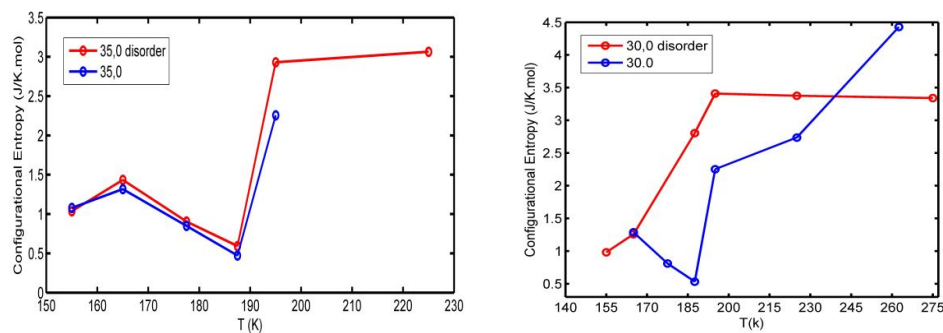


Fig.13

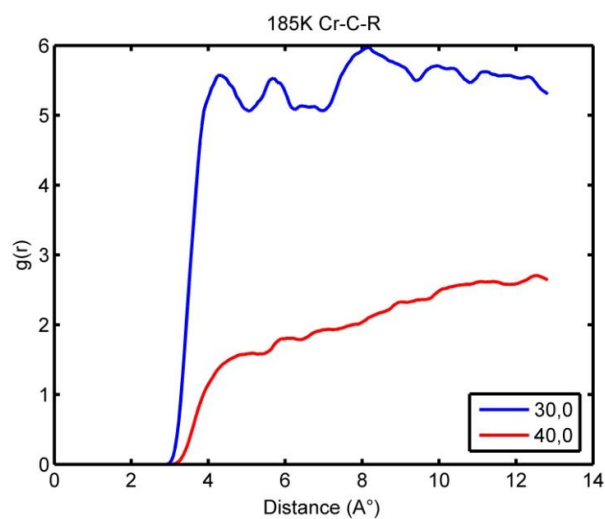


Fig.14.

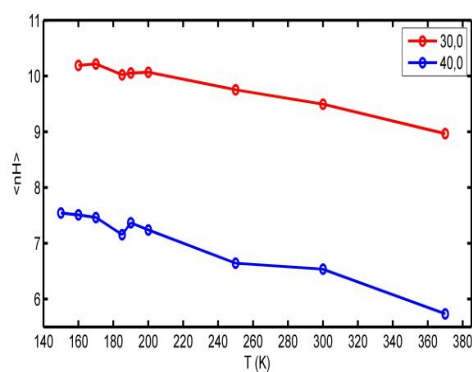


Fig.15.

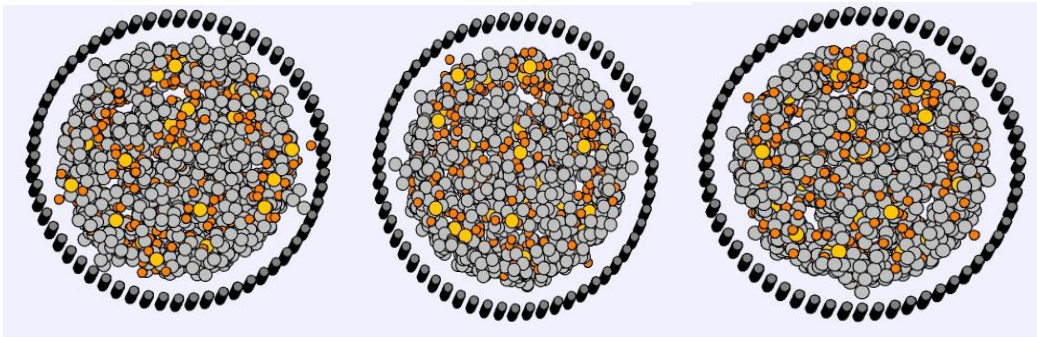
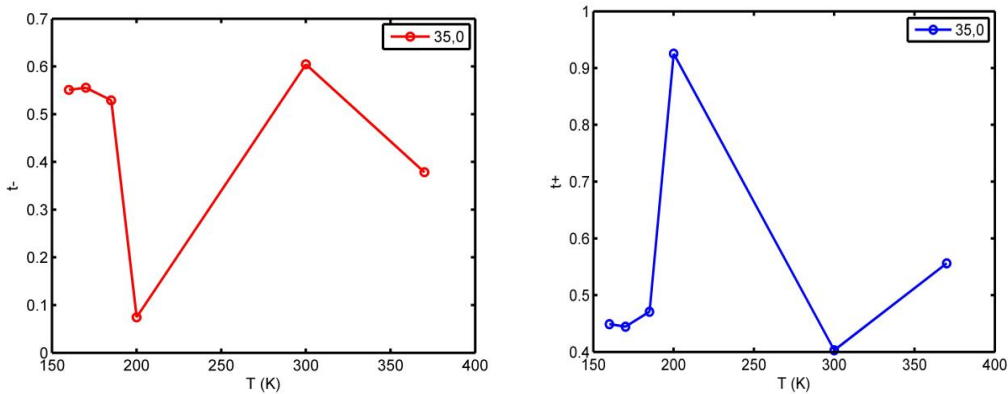
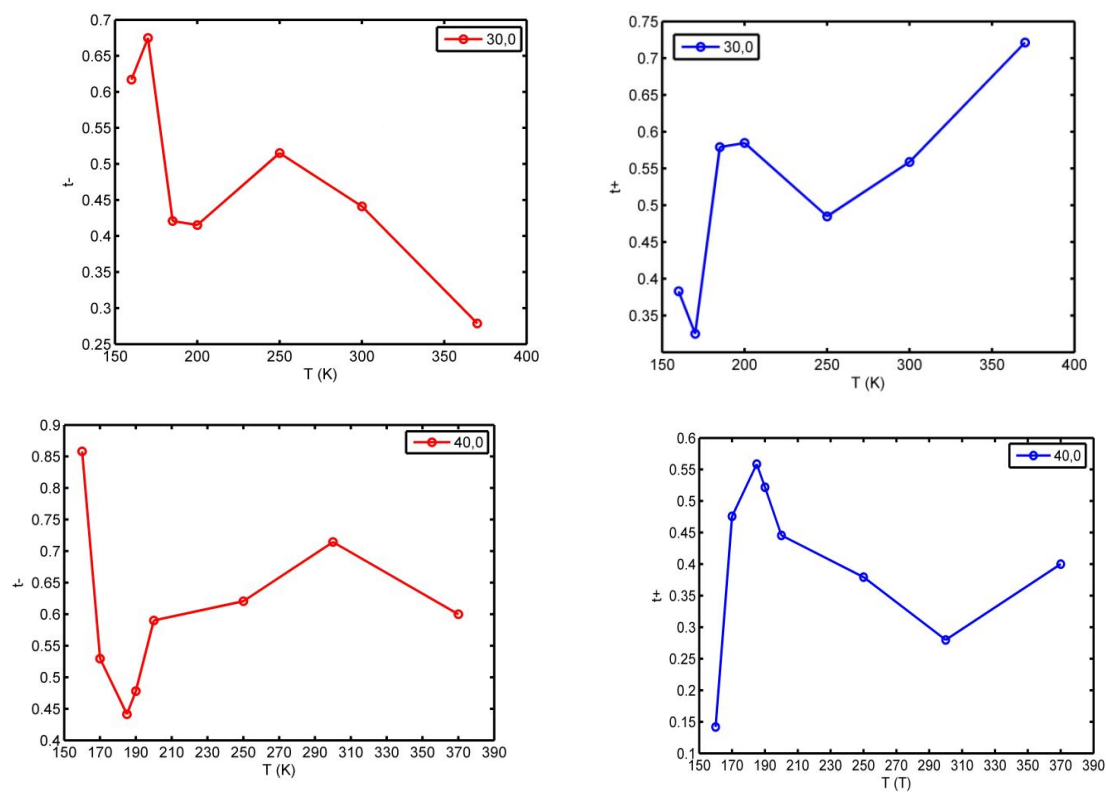
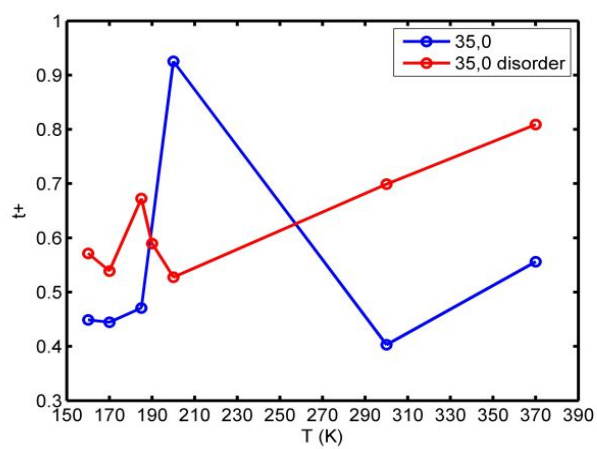
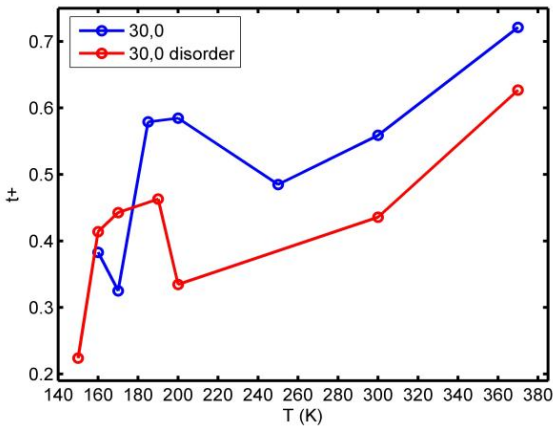


Fig.16

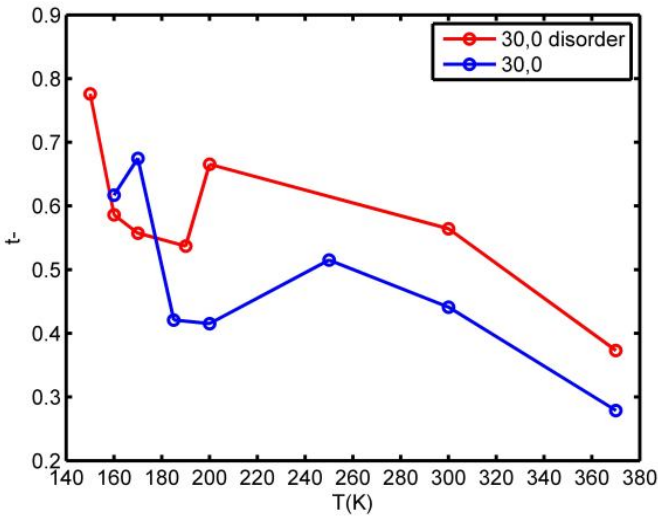


**Fig.17.****a)**



b)

Fig.18



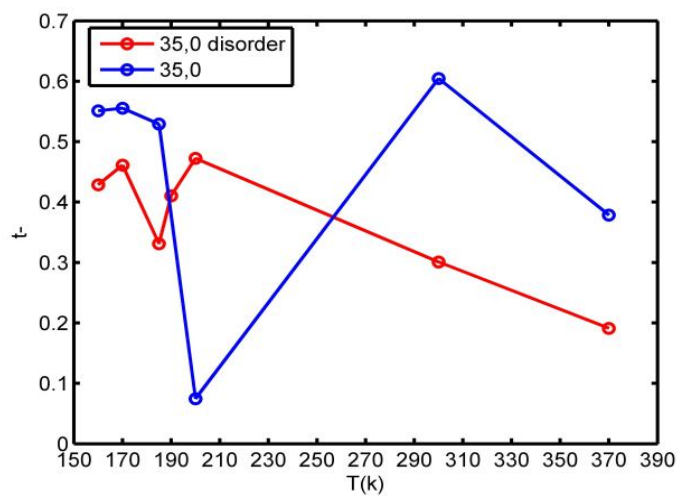
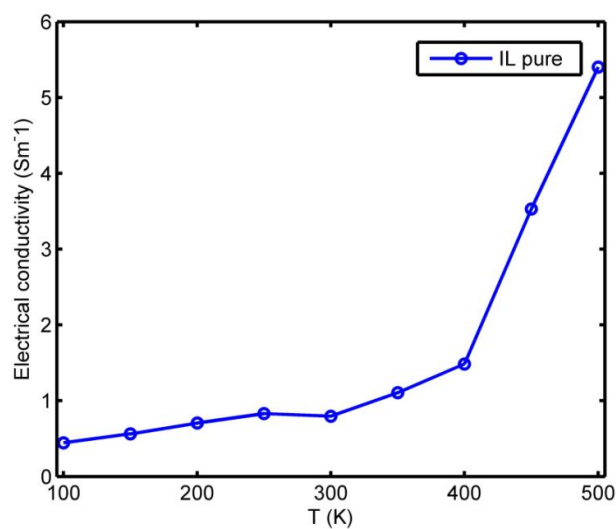
**Fig.19**

Fig.20

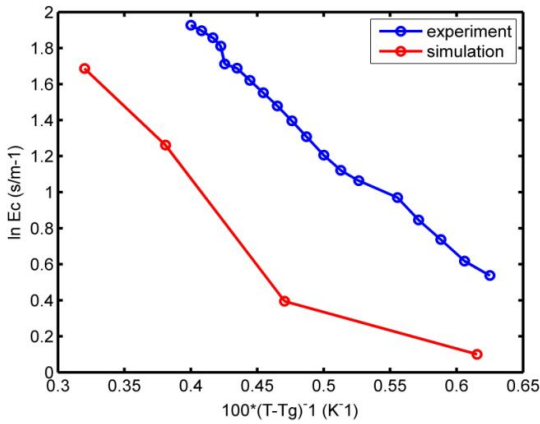


Fig.21.a

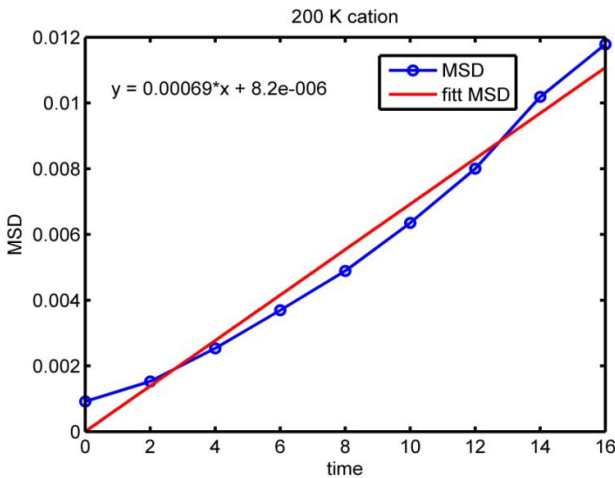


Fig.21.b

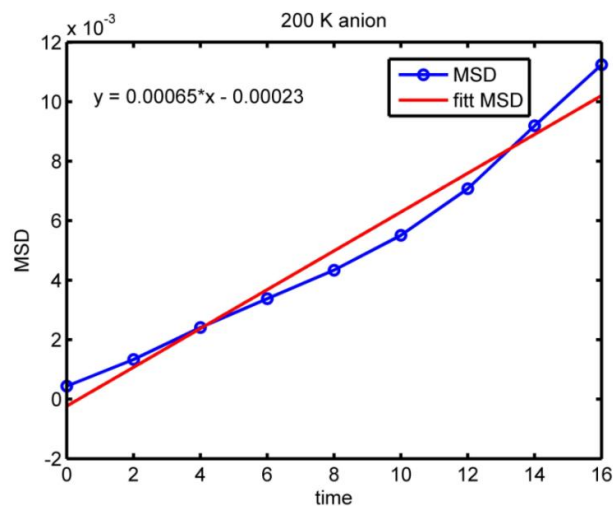


Fig.22

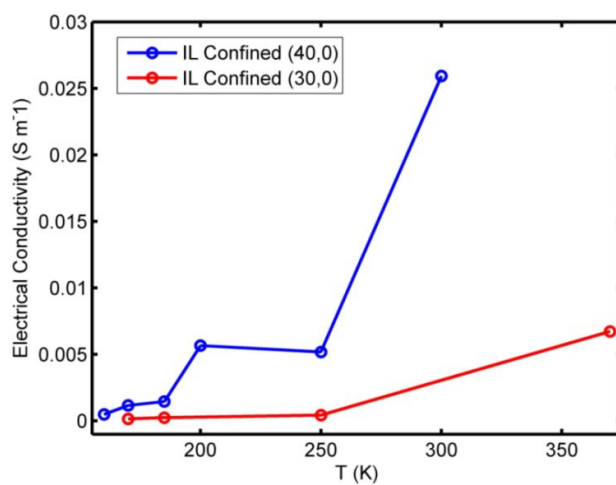


Fig.23

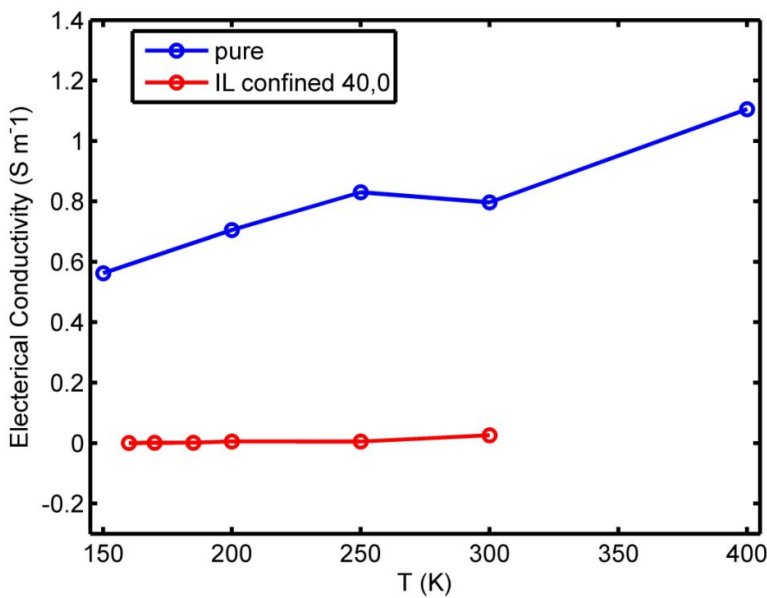
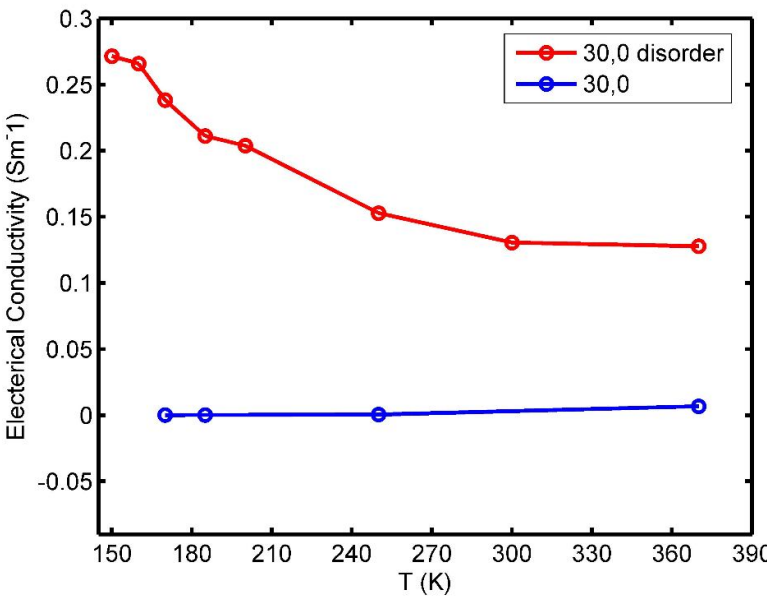
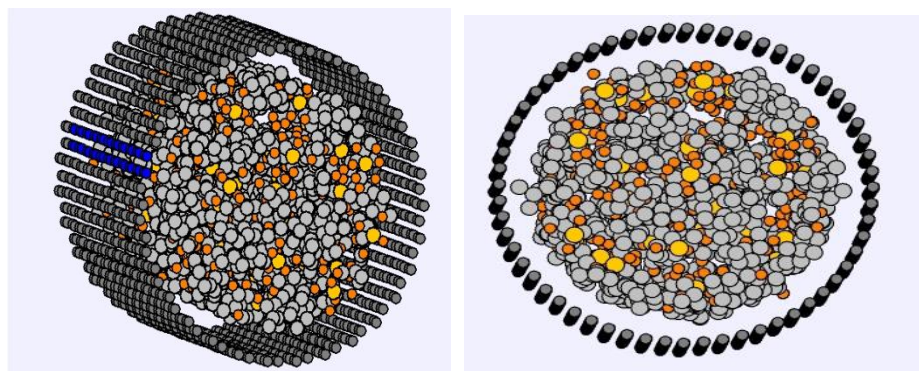
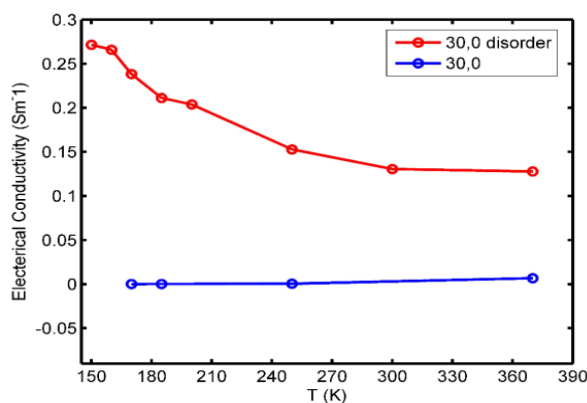


Fig.24.





Schematic presentation of [EMIM][PF6] encapsulated in zigzag type of carbon nanotube in presence nitrogen doping.



Electrical conductivity of [EMIM][PF6] increases versus the temperature when it encapsulated in CNT however electrical conductivity trend is reversed with temperature when nitrogen doped in zigzag CNT. In presence of impurity in CNT, ion transfer has been done via hydrogen bonding mechanism however in absence impurity; ion transfer has been done via diffusion mechanism in which cation has significant effect on ion transfer and electrical conductivity.